Multiple-Choice Questions on Program Design and Analysis

1. A program that reads in a five-digit identification number is to be written. The specification does not state whether zero can be entered as a first digit. The programmer should

   (A) write the code to accept zero as a first digit since zero is a valid digit.
   (B) write the code to reject zero as a first digit since five-digit integers do not start with zero.
   (C) treat the identification number as a four-digit number if the user enters a number starting with zero.
   (D) check with the writer of the specification whether zero is acceptable as a first digit.

2. Refer to the following three program descriptions.

   I. Test whether there exists at least one three-digit integer whose value equals the sum of the squares of its digits.
   II. Read in a three-digit code number and check if it is valid according to some given formula.
   III. Passwords consist of three digits and three capital letters in any order. Read in a password, and check if there are any repeated characters.

   For which of the preceding program descriptions would a `ThreeDigitNumber` class be suitable?

   (A) I only
   (B) III only
   (C) I and II only
   (D) I, II, and III

3. Top-down programming is illustrated by which of the following?

   (A) Writing a program from top to bottom in Java
   (B) Using driver programs to test all methods in the order that they're called in the program
   (C) Writing and testing the lowest level methods first and then combining them to form appropriate abstract operations
   (D) Writing the program in terms of the operations to be performed and then refining these operations by adding more detail

4. Which of the following should influence your choice of a particular algorithm?

   I. The run time of the algorithm
   II. The memory requirements of the algorithm
   III. The ease with which the logic of the algorithm can be understood

   (A) I only
   (B) I and III only
   (C) I and II only
   (D) I, II, and III

5. A list of numbers is stored in a sorted array. It is required that the list be maintained in sorted order. This requirement leads to inefficient execution for which of the following processes?

   I. Summing the five smallest numbers in the list
   II. Finding the maximum value in the list
   III. Inserting and deleting numbers

   (A) I only
   (B) III only
   (C) II and III only
   (D) I and III only

6. Which of the following is not necessarily a feature of a robust program?

   (A) Does not allow execution to proceed with invalid data
   (B) Uses algorithms that give correct answers for extreme data values
   (C) Will run on any computer without modification
   (D) Will anticipate the types of errors that users of the program may make

7. A certain freight company charges its customers for shipping overseas according to this scale.

   $80 per ton for a weight of 10 tons or less
   $40 per ton for each additional ton over 10 tons but not exceeding 25 tons
   $30 per ton for each additional ton over 25 tons

   For example, to ship a weight of 12 tons will cost 10(80) + 2(40) = $880. To ship 26 tons will cost 10(80) + 15(40) + 1(30) = $1430.
   A method takes as parameter an integer that represents a valid shipping weight and outputs the charge for the shipment. Which of the following is the smallest set of input values for shipping weights that will adequately test this method?

   (A) 10, 25
   (B) 5, 15, 30
   (C) 5, 10, 15, 25, 30
   (D) 0, 5, 10, 15, 25, 30

8. A code segment calculates the mean of values stored in integers `n1`, `n2`, `n3`, and `n4` and stores the result in `average`, which is of type `double`. What kind of error is caused with this statement?

```
double average = n1 + n2 + n3 + n4
    / (double) 4;
```

   (A) Logic
   (B) Run-time
   (C) Syntax
   (D) Type mismatch

9. A program evaluates binary arithmetic expressions that are read from an input file. All of the operands are integers, and the only operators are +, -, *, and /. In writing the program, the programmer forgot to include a test that checks whether the right-hand operand in a division expression equals zero. When will this oversight be detected by the computer?

   (A) At compile time
   (B) As soon as the data from the input file is read
   (C) During evaluation of the expressions
   (D) When at least one incorrect value for the expressions is output

10. A programmer plans to write a program that simulates various games. In the program, there is a `Player` class that has a `getMove` method. Method `getMove` returns an `int` value to simulate a move in a game. Which of the games described below are suitable candidates for using the `getMove` method as specified above?

    I. High-Low Guessing Game: The computer thinks of a number and the player who guesses it with the least number of guesses wins. After each guess, the computer tells whether its number is higher or lower than the player's guess.
    II. Chips: Start with a pile of chips. Each player, in turn, removes some number of chips, but not all of them. The winner is the one who removes the final chip.
    III. Tic-Tac-Toe: Two players alternate placing "X" or "O" on a 3 × 3 grid. The first player to get three in a row, where a row can be horizontal, vertical, or diagonal, wins.

    (A) I only
    (B) III only
    (C) I and II only
    (D) I, II, and III

11. Consider the following code fragment.

```
/** Precondition: a1, a2, a3 contain 3
*     distinct integers.
* Postcondition: max contains the
* largest of a1, a2, a3.
*/
//first set max equal to larger
//    of a1 and a2
if (a1 > a2)
    max = a1;
else
    max = a2;
//set max equal to larger of max
// and a3
if (max < a3)
    max = a3;
```

For this algorithm, which of the following initial setups for `a1`, `a2`, and `a3` will cause
(1) the least number of computer operations (best case) and
(2) the greatest number of computer operations (worst case)?

(A) (1) largest value in `a1` or `a2`
(2) largest value in `a3`

(B) (1) largest value in `a2` or `a3`
(2) largest value in `a1`

(C) (1) smallest value in `a1`
(2) largest value in `a2`

(D) (1) largest value in `a2`
(2) smallest value in `a3`

12. The sequence of Fibonacci numbers is 1, 1, 2, 3, 5, 8, 13, 21, ... The first two Fibonacci numbers are each 1. Each subsequent number is obtained by adding the previous two. Consider this method.

```
/** Precondition: n >= 1.
 * Postcondition: The nth Fibonacci
 * number has been returned.
 */
public static int fib(int n)
{
    int prev = 1, next = 1, sum = 1;
    for (int i = 3; i <= n; i++)
    {
        /* assertion */
        sum = next + prev;
        prev = next;
        next = sum;
    }
    return sum;
}
```

Which of the following is a correct `/* assertion */` about the loop variable `i`?

(A) $1 \le i \le n$
(B) $0 \le i \le n$
(C) $3 \le i \le n$
(D) $3 < i < n+1$

13. A program is to be written that prints an invoice for a small store. A copy of the invoice will be given to the customer and will display:

- A list of items purchased.
- The quantity, unit price, and total price for each item.
- The amount due.

Three candidate classes for this program are `Invoice`, `Item`, and `ItemList`, where an `Item` is a single item purchased and `ItemList` is the list of all items purchased. Which class is a reasonable choice to be responsible for the `amountDue` method, which returns the amount the customer must pay?

I. `Item`
II. `ItemList`
III. `Invoice`

(A) I only
(B) III only
(C) I and II only
(D) II and III only

14. Which is a false statement about classes in object-oriented program design?

(A) If a class `C1` has an instance variable whose type is another class, `C2`, then `C1` *has-a* `C2`.
(B) If a class `C1` is associated with another class, `C2`, then `C1` depends on `C2` for its implementation.
(C) If classes `C1` and `C2` are related such that `C1` *is-a* `C2`, then `C2` *has-a* `C1`.
(D) If class `C1` is independent, then none of its methods will have parameters that are objects of other classes.

15. A Java programmer has completed a preliminary design for a large program. The programmer has developed a list of classes, determined the methods for each class, established the relationships between classes, and written an outline for each class. Which class(es) should be implemented first?

(A) Any superclasses
(B) Any subclasses
(C) All collaborator classes (classes that will be used to implement other classes)
(D) All independent classes (classes that have no references to other classes)

Use the program description below for Questions 16–18.

A program is to be written that simulates bumper cars in a video game. The cars move on a square grid and are located on grid points (x,y), where x and y are integers between -20 and 20. A bumper car moves in a random direction, either left, right, up, or down. If it reaches a boundary (i.e., x or y is ± 20), then it reverses direction. If it is about to collide with another bumper car, it reverses direction. Your program should be able to add bumper cars and run the simulation. One step of the simulation allows each car in the grid to move. After a bumper car has reversed direction twice, its turn is over and the next car gets to move.

16. To identify classes in the program, the nouns in the specification are listed:

program, bumper car, grid, grid point, integer, direction, boundary, simulation

How many nouns in the list should immediately be discarded because they are unsuitable as classes for the program?

(A) 0
(B) 1
(C) 2
(D) 3

A programmer decides to include the following classes in the program. Refer to them for Questions 17 and 18.

- `Simulation` will run the simulation.
- `Display` will show the state of the game.
- `BumperCar` will know its identification number, position in the grid, and current direction when moving.
- `GridPoint` will be a position in the grid. It will be represented by two integer fields, `x_coord` and `y_coord`.
- `Grid` will keep track of all bumper cars in the game, the number of cars, and their positions in the grid. It will update the grid each time a car moves. It will be implemented with a 2D array of `BumperCar`.

17. Which operation should not be the responsibility of the `GridPoint` class?

(A)	`isEmpty`	returns false if the grid point contains a `BumperCar`, true otherwise
(B)	`atBoundary`	returns true if x or y coordinate = ± 20, false otherwise
(C)	`left`	if not at left boundary, change the grid point to 1 unit left of current point
(D)	`get_x`	return x-coordinate of this point

18. Which method is not suitable for the `BumperCar` class?

(A)
```
public boolean atBoundary()
//Returns true if BumperCar at boundary, false otherwise.
```

(B)
```
public void selectRandomDirection()
//Select random direction (up, down, left, or right)
// at start of turn.
```

(C)
```
public void reverseDirection()
//Move to grid position that is in direction opposite to
// current direction.
```

(D)
```
public void update()
//Modify Grid to reflect new position after each stage
// of move.
```